

Affect x conceptual information in the battle of linguistic meaning

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Affect-First Hypothesis

Affective Info takes precedence over **conceptual** Info [1,2].



Question 1

Does the AFH hold across the lexicon?

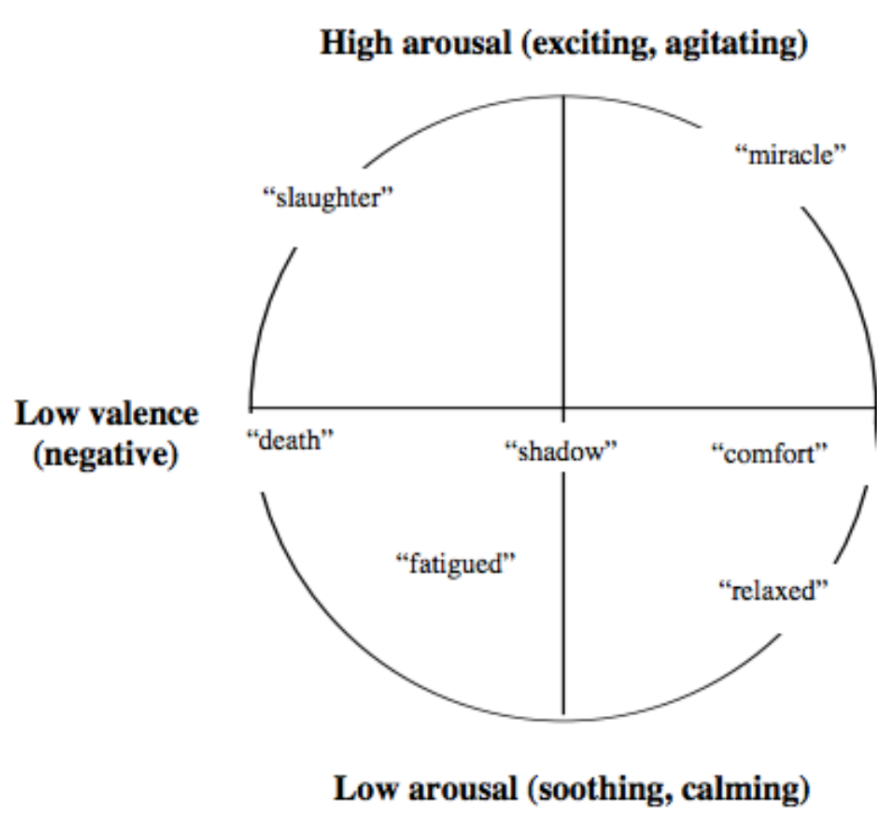
Question 2a

Is all Conceptual Info the same compared to affect?

Question 2b

How does Conceptual Info relate to (core) cognition?

Affect/Valence: Ling & Psych Background



Semantics Affect/expressive/evaluative meaning non-primary dimension following Frege, though challenged more and more [e.g., 3-9]

Psychology All concepts located in affective space [10-16], BUT **status of AFH still Empirically Murky**, for: [17-26], against: [27-35]

• Sensitive to task [36-38], conversational context/topicality (people) [39-40] • Informative about speaker's meaning/referential intentions [41-42] •

Conceptual Foils Find conceptual dimensions as cognitively similar to Affect as possible, i.e. that play **critical roles in cognition, development, language**.

• **Concrete/Abstract** (Exps 1a-1c) [44-54] • **Physical vs. Psychological/Social** (Exps 2a-2b) [65-70] • **Animacy** (Exp 3) [71-78] • **Syntax** (Exp 4) [e.g., 79] •

Exps 1-4: Speeded Categorization (n=270)



progress

mud

F

J

concrete

abstract

- Words normed for Valence x Conceptual feature combo [eg. 15,82-83]
- Speeded: participants respond AFAP
- **ConceptualFoil** between-subjs
- **Task** 2x1 within-subjs design:
 - **Valence**, **Conceptual**
 - Blocked, order counterbalanced
 - Words presented 3x per block

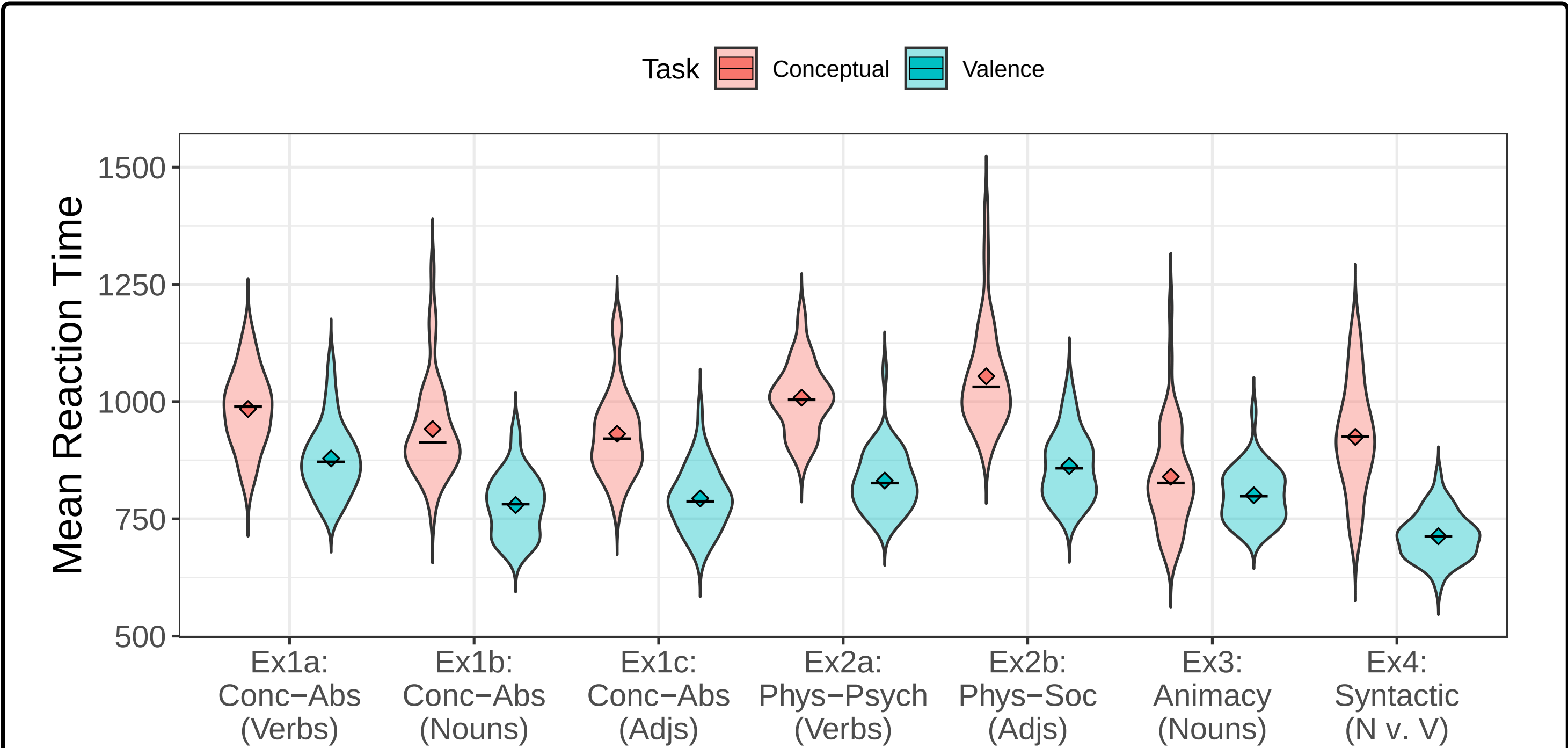


Figure 1: **Valence** < **Conceptual** except **Animacy** (Ex3).

Analysis & Results

$\text{lmer}(\text{LogRT} \sim \text{Task} + (1 + \text{Task} | \text{ID}) + (1 + \text{Task} | \text{Word}))$

Valence < **Conceptual** except **Animacy**, but Valence still faster mean RT.

- **Q:** Are Conceptual Foils slower because of **uncertainty** about meaning?
→ **No**, because Affect faster even on most accurate(=unambig.) words, Fig. 2.
- **Q:** Is Affect faster because we're in the **habit** of making these judgments?
→ **No**, because why not greater advantage over Animacy, which is not habitual?

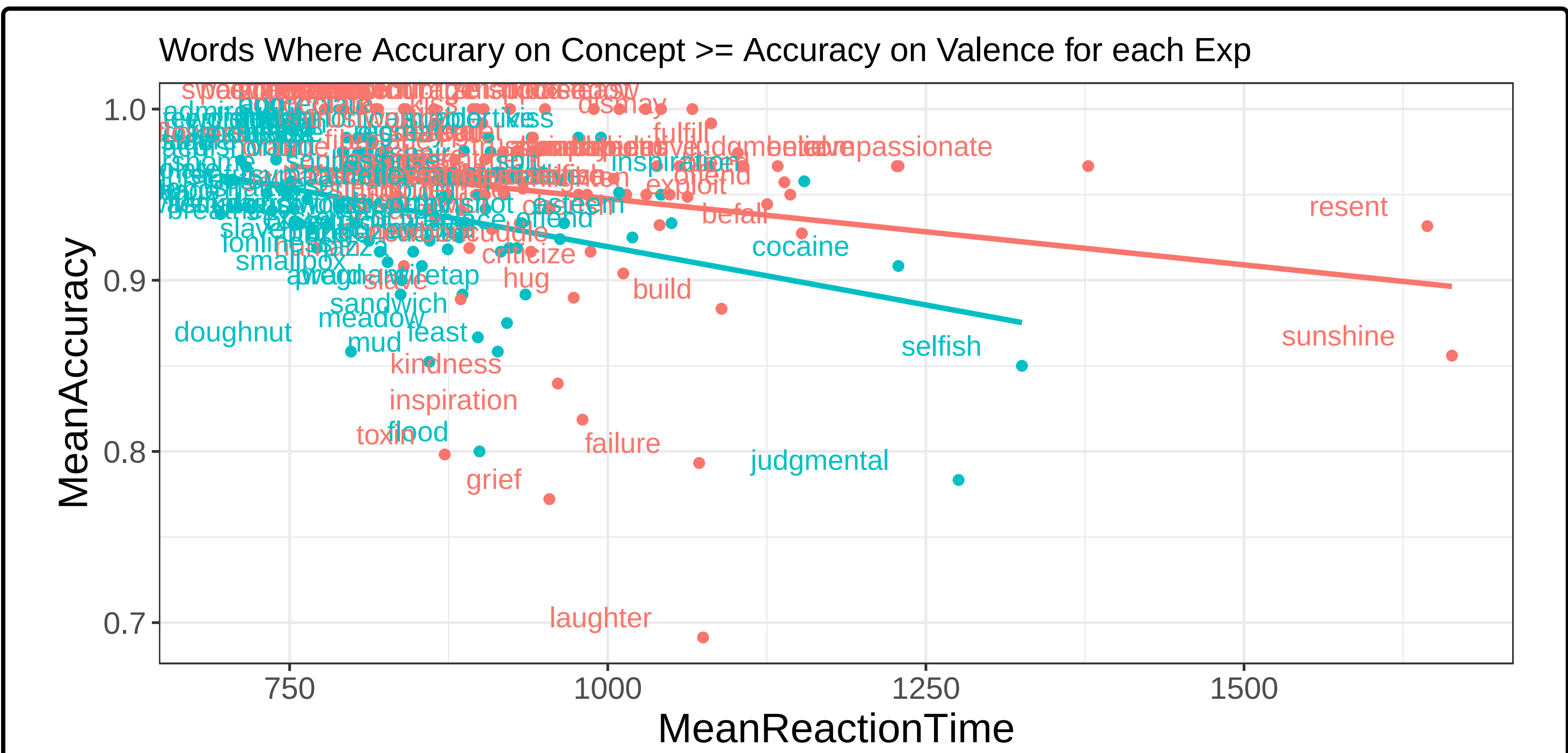


Figure 2: No strong RT~Accuracy relation in high agreement words.

Exps 5-8: Similarity Judgments (n=350)



progress

cuddle

corrode

How similar is the word in red to the word in green?

completely dissimilar

completely similar

Next

- **ConceptualFoil** between-subjs
- Items = WordPairs from Exps 1-7
- Items presented in randomized order
- **FeatureMatch**: 4 x 1 within-subjs
 - **Test**: **ValenceMatchingOnly**, **ConceptualMatchingOnly**, **MaxMismatch**, **MaxMatch**
 - **DissimilarControls** (Antonyms), **SimilarControls** (Synonyms)

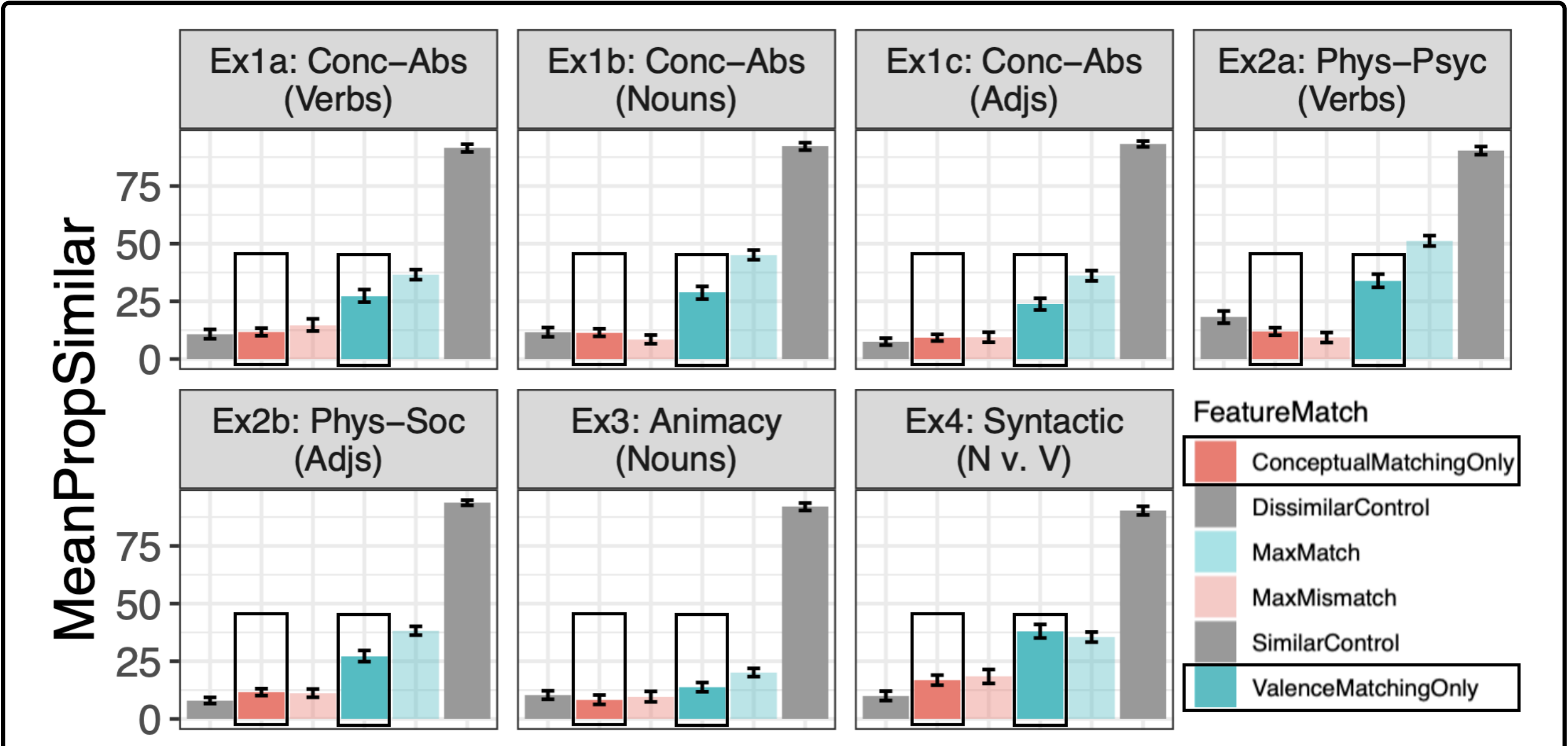


Figure 3: **ValenceMatch** pairs more similar than **ConceptualMatch**.

Analysis & Results

$\text{lmer}(\text{Rating} \sim \text{FeatureMatch} + (1 + \text{FeatureMatch} | \text{ID}) + (1 | \text{Word}))$

Valence drives judgments of similarity more than any Conceptual Foil

- **ValenceMatchingOnly** pairs **more similar** than any **ConceptualMatchingOnly**.
- **ConceptualMatchingOnly** pairs no more similar than **MaxMismatch** pairs.

Discussion and Conclusion

• **Affective dimension drives meaning in isolated words**

- Exps 1-7: Valence categorization faster than conceptual categorization; except Animacy
- Exps 8-14: Similarity judged more based on Matching Valence and not Matching Conceptual
- Patterns consistent across items tested, but subgroups emerge

• Step towards understanding how valence meaning can change in sentential & discourse contexts [36-42].

• **Probabilistic pragmatics**: build up sentential & discourse contexts to understand interacting info [e.g., 87].

• **Psychosemantics**: meanings “point to” concepts [84-86]; affective meanings→affective cognition; would explain Animacy too.

References [1] Wundt 1907 [2] Zajonc 1980, 2000 [3] Potts 2005 [4] Schlenker 2007 [5] Kaplan 1999 [6] McCready 2010, 2020 [7] Gutzmann 2015 [8] Stojanovic & Kaiser 2023 [9] Stojanovic in press [10] Osgood et al 1957 [11] Mehrabian et Russell 1974 [12] Lambon Ralph et al 2017 [13] Yap & Show 2017 [14] Bradley & Lang 1999 [15] Warriner et al. 2013 [16] Mohammad 2018 [17] Zajonc 1968 [18] Fazio 2001 [19] Costa et al 2009 [20] Vigliocco et al 2011 [21] Scott et al 2009 [22] Scott et al 2012 [23] Bayer et al 2012 [24] Rellecke et al 2011 [25] Palazova et al 2011 [26] Hinojosa et al 2018, 2019, 2020, 2023 [27] Lazarus 1984 [28] Rolls 1999 [29] Storbeck et al 2006 [30] Storbeck & Clore 2007 [31] Calvo & Nummenmaa 2007 [32] Batty, et al 2005 [33] Fox et al., 2000 [34] Horstmann & Bauland, 2006 [35] Nummenmaa et al 2010 [36] Hinojosa et al 2014 [37] Hinojosa et al 2019 [38] Kaltwasser et al 2013 [39] Rohr & Wentura 2022 [40] Gotzner & Mazzarella 2024 [41] Mazzarella & Gotzner 2021 [42] Ronderos & Domenech 2023 [43] Paivio 1971 [44] James 1975 [45] Rudin 1980 [46] Paivio 1986 [47] Kroll et al 1986 [48] Schwanenflugel et al 1988 [49] Binder et al 2005 [50] Palazova et al 2003 [51] Herbert et al [52] Kansuke & Kotz 2007 [53] Kousta et al 2011 [54] Meteyard et al 2012 [55] Vigliocco et al 2014 [56] Borghi et al 2017, 2018, 2019 [57] Ponari et al, 2018 [58] Barsalou et al 2018 [59] Paulig et al 2019 [60] Hinojosa et al 2019 [61] Lund et al 2019 [62] Kim et al 2020 [63] Winter 2022 [64] Strickland et al. 2017 [65] Kuhn et al. 2021 [66] Leslie & Keeble 1987 [67] Spelke et al. 1995 [68] Saxe et al. 2005 [69] Newman et al. 2010 [70] Rakison & Poulin-Dubois 2001 [71] Caramazza & Shelton 1998 [72] Spelke, Phillips, & Woodward 1995 [73] Childers & Echols 2004 [74] Baron-Cohen 1995 [75] Carey 1985 [76] Vihman & Nelson 2019 [77] Haspelmath 2013 [78] Strijkers & Costa 2011 [79] Brysbaert et al 2013 [80] Wordnet: Fellbaum (1998) [81] Pietroski (2018) [82] Pietroski (2025) [83] Lidz, Pietroski, Halberda, Hunter (2011) [84] Degen & Tanenhaus (2019)

Thank you!

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